

Services/MySQLCoordinatorRepository.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using MySql.Data.MySqlClient;
5
6 namespace Week3
7 {
8     public class MySQLCoordinatorRepository : ICoordinatorRepository, IDisposable
9     {
10         private readonly string _connectionString;
11         private MySqlConnection? _conn;
12
13         public MySQLCoordinatorRepository(string connectionString)
14         {
15             _connectionString = connectionString ?? throw new ArgumentNullException(nameof(connectionString));
16             _conn = new MySqlConnection(_connectionString);
17             _conn.Open();
18             EnsureSchema();
19         }
20
21         private void EnsureSchema()
22         {
23             // Drop dependent tables first if they exist (to handle schema updates)
24             var dropSql = @"
25 DROP TABLE IF EXISTS assignments;
26 DROP TABLE IF EXISTS internship_assignments;
27 DROP TABLE IF EXISTS internship_contacts;
28 DROP TABLE IF EXISTS internships;
29 DROP TABLE IF EXISTS contact_persons;
30 DROP TABLE IF EXISTS students;
31 DROP TABLE IF EXISTS organizations;
32 ";
33             using (var dropCmd = _conn!.CreateCommand())
34             {
35                 dropCmd.CommandText = dropSql;
36                 dropCmd.ExecuteNonQuery();
37             }
38
39             var sql = @"
40 CREATE TABLE organizations (
41     organization_id VARCHAR(36) PRIMARY KEY,
42     name VARCHAR(200) NOT NULL,
43     address VARCHAR(500),
44     city VARCHAR(100),
45     url VARCHAR(500),
46     phone VARCHAR(50),
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47     email VARCHAR(200)
48 );
49
50 CREATE TABLE contact_persons (
51     id VARCHAR(36) PRIMARY KEY,
52     organization_id VARCHAR(36),
53     first_name VARCHAR(100),
54     last_name VARCHAR(100),
55     email VARCHAR(200),
56     job_title VARCHAR(200),
57     phone VARCHAR(50),
58     department VARCHAR(200),
59     CONSTRAINT FK_contact_org FOREIGN KEY (organization_id) REFERENCES
60     organizations(organization_id) ON DELETE CASCADE
61 );
62
63 CREATE TABLE internships (
64     id VARCHAR(36) PRIMARY KEY,
65     organization_id VARCHAR(36),
66     title VARCHAR(500),
67     short_description TEXT,
68     long_description TEXT,
69     year INT,
70     semester INT,
71     category INT,
72     is_completed INT(1) DEFAULT 0,
73     final_grade DOUBLE NULL,
74     CONSTRAINT FK_intern_org FOREIGN KEY (organization_id) REFERENCES
75     organizations(organization_id) ON DELETE CASCADE
76 );
77
78 CREATE TABLE internship_contacts (
79     internship_id VARCHAR(36),
80     contact_id VARCHAR(36),
81     PRIMARY KEY (internship_id, contact_id),
82     CONSTRAINT FK_ic_intern FOREIGN KEY (internship_id) REFERENCES internships(id) ON
83     DELETE NO ACTION,
84     CONSTRAINT FK_ic_contact FOREIGN KEY (contact_id) REFERENCES contact_persons(id)
85     ON DELETE NO ACTION
86 );
87
88 CREATE TABLE students (
89     student_id VARCHAR(50) PRIMARY KEY,
90     first_name VARCHAR(200),
91     last_name VARCHAR(200)
92 );
93
94 CREATE TABLE internship_assignments (
95     internship_id VARCHAR(36),
96     student_id VARCHAR(50),

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93     PRIMARY KEY (internship_id, student_id),
94     CONSTRAINT FK_assign_intern FOREIGN KEY (internship_id) REFERENCES
internships(id) ON DELETE CASCADE,
95     CONSTRAINT FK_assign_student FOREIGN KEY (student_id) REFERENCES
students(student_id) ON DELETE CASCADE
96 );
97
98 CREATE TABLE assignments (
99     assignment_id VARCHAR(36) PRIMARY KEY,
100    internship_id VARCHAR(36),
101    title VARCHAR(500) NOT NULL,
102    assignment_type VARCHAR(50),
103    minor_module VARCHAR(200),
104    is_completed INT(1) DEFAULT 0,
105    CONSTRAINT FK_assign_intern_2 FOREIGN KEY (internship_id) REFERENCES
internships(id) ON DELETE CASCADE
106 );
107 ";
108         using var createCmd = _conn!.CreateCommand();
109         createCmd.CommandText = sql;
110         createCmd.ExecuteNonQuery();
111     }
112
113     private static string NewId() => Guid.NewGuid().ToString();
114
115     public IReadOnlyList<Organization> Organizations => GetOrganizationsOver-
view().ToList();
116     public IReadOnlyList<Internship> Internships =>
GetAllInternships().ToList();
117     public IReadOnlyList<Student> Students => GetAllStudents().ToList();
118
119     public bool AddOrganization(Organization org)
120     {
121         if (org == null) return false;
122
123         using var check = _conn!.CreateCommand();
124         check.CommandText = "SELECT COUNT(*) FROM organizations WHERE
LOWER(name)=LOWER(@name) AND LOWER(city)=LOWER(@city)";
125         check.Parameters.AddWithValue("@name", org.Name);
126         check.Parameters.AddWithValue("@city", org.City);
127         var exists = Convert.ToInt32(check.ExecuteScalar() ?? 0) > 0;
128         if (exists) return false;
129
130         using var cmd = _conn.CreateCommand();
131         cmd.CommandText = @"INSERT INTO organizations (organization_id, name,
address, city, url, phone, email)
132 VALUES (@id,@name,@address,@city,@url,@phone,@email)";
133         cmd.Parameters.AddWithValue("@id", org.OrganizationId);
134         cmd.Parameters.AddWithValue("@name", org.Name);
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135         cmd.Parameters.AddWithValue("@address", org.Address ??  
(object)DBNull.Value);  
136         cmd.Parameters.AddWithValue("@city", org.City ?? (object)DBNull.Value);  
137         cmd.Parameters.AddWithValue("@url", org.Url ?? (object)DBNull.Value);  
138         cmd.Parameters.AddWithValue("@phone", org.PhoneNumber ??  
(object)DBNull.Value);  
139         cmd.Parameters.AddWithValue("@email", org.Email ??  
(object)DBNull.Value);  
140         cmd.ExecuteNonQuery();  
141  
142         foreach (var c in org.Contacts)  
143         {  
144             using var ccmd = _conn.CreateCommand();  
145             ccmd.CommandText = @"  
146 INSERT INTO contact_persons (id, organization_id, first_name, last_name, email,  
job_title, phone, department)  
147 VALUES (@id,@org,@first,@last,@email,@job,@phone,@dept)  
148 ON DUPLICATE KEY UPDATE first_name=VALUES(first_name), last_name=VALUES(last_name),  
email=VALUES(email), job_title=VALUES(job_title), phone=VALUES(phone),  
department=VALUES(department), organization_id=VALUES(organization_id);";  
149             ccmd.Parameters.AddWithValue("@id", c.Id);  
150             ccmd.Parameters.AddWithValue("@org", org.OrganizationId);  
151             ccmd.Parameters.AddWithValue("@first", c.FirstName ??  
(object)DBNull.Value);  
152             ccmd.Parameters.AddWithValue("@last", c.LastName ??  
(object)DBNull.Value);  
153             ccmd.Parameters.AddWithValue("@email", c.Email ??  
(object)DBNull.Value);  
154             ccmd.Parameters.AddWithValue("@job", c.JobTitle ??  
(object)DBNull.Value);  
155             ccmd.Parameters.AddWithValue("@phone", c.PhoneNumber ??  
(object)DBNull.Value);  
156             ccmd.Parameters.AddWithValue("@dept", c.Department ??  
(object)DBNull.Value);  
157             ccmd.ExecuteNonQuery();  
158         }  
159  
160         return true;  
161     }  
162  
163     public IEnumerable<Organization> GetOrganizationsOverview()  
164     {  
165         var list = new List<Organization>();  
166         using var cmd = _conn!.CreateCommand();  
167         cmd.CommandText = "SELECT organization_id, name, address, city, url,  
phone, email FROM organizations ORDER BY city, name";  
168         using var rdr = cmd.ExecuteReader();  
169         while (rdr.Read())  
170         {  
171             var org = new Organization(  
172                 rdr.GetString("organization_id"),
```

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173         rdr.GetString("name"),
174         rdr.IsDBNull("address") ? "" : rdr.GetString("address"),
175         rdr.IsDBNull("city") ? "" : rdr.GetString("city")
176     )
177     {
178         Url = rdr.IsDBNull("url") ? null : rdr.GetString("url"),
179         PhoneNumber = rdr.IsDBNull("phone") ? null :
rdr.GetString("phone"),
180         Email = rdr.IsDBNull("email") ? null : rdr.GetString("email")
181     };
182     list.Add(org);
183 }
184 rdr.Close();
185
186 foreach (var org in list)
187 {
188     using var ccmd = _conn.CreateCommand();
189     ccmd.CommandText = "SELECT id, first_name, last_name, email,
job_title, phone, department FROM contact_persons WHERE organization_id=@org ORDER
BY last_name, first_name";
190     ccmd.Parameters.AddWithValue("@org", org.OrganizationId);
191     using var cr = ccmd.ExecuteReader();
192     while (cr.Read())
193     {
194         var cp = new ContactPerson
195         {
196             Id = cr.GetString("id"),
197             FirstName = cr.IsDBNull("first_name") ? null :
cr.GetString("first_name"),
198             LastName = cr.IsDBNull("last_name") ? null :
cr.GetString("last_name"),
199             Email = cr.IsDBNull("email") ? null :
cr.GetString("email"),
200             JobTitle = cr.IsDBNull("job_title") ? null :
cr.GetString("job_title"),
201             PhoneNumber = cr.IsDBNull("phone") ? null :
cr.GetString("phone"),
202             Department = cr.IsDBNull("department") ? null :
cr.GetString("department")
203         };
204         org.Contacts.Add(cp);
205     }
206     cr.Close();
207 }
208
209 return list;
210 }
211
212 public void AddInternship(Internship internship)
213 {
```

```
214         using var cmd = _conn!.CreateCommand();
215         cmd.CommandText = @"INSERT INTO internships (id, organization_id,
title, short_description, long_description, year, semester, category, is_completed,
final_grade)
216 VALUES (@id,@org,@title,@short,@long,@year,@sem,@cat,0,NULL)";
217         cmd.Parameters.AddWithValue("@id", internship.Id);
218         cmd.Parameters.AddWithValue("@org", internship.OrganizationId);
219         cmd.Parameters.AddWithValue("@title", internship.Title);
220         cmd.Parameters.AddWithValue("@short", internship.ShortDescription ??
(object)DBNull.Value);
221         cmd.Parameters.AddWithValue("@long", internship.LongDescription ??
(object)DBNull.Value);
222         cmd.Parameters.AddWithValue("@year", internship.Period.Year);
223         cmd.Parameters.AddWithValue("@sem", internship.Period.Semester);
224         cmd.Parameters.AddWithValue("@cat", (int)internship.Category);
225         cmd.ExecuteNonQuery();
226
227         foreach (var c in internship.Contacts)
228         {
229             using var ccmd = _conn.CreateCommand();
230             ccmd.CommandText = @"
231 INSERT INTO contact_persons (id, organization_id, first_name, last_name, email,
job_title, phone, department)
232 VALUES (@id,@org,@first,@last,@email,@job,@phone,@dept)
233 ON DUPLICATE KEY UPDATE first_name=VALUES(first_name), last_name=VALUES(last_name),
email=VALUES(email), job_title=VALUES(job_title), phone=VALUES(phone),
department=VALUES(department), organization_id=VALUES(organization_id);";
234             ccmd.Parameters.AddWithValue("@id", c.Id);
235             ccmd.Parameters.AddWithValue("@org", internship.OrganizationId);
236             ccmd.Parameters.AddWithValue("@first", c.FirstName ??
(object)DBNull.Value);
237             ccmd.Parameters.AddWithValue("@last", c.LastName ??
(object)DBNull.Value);
238             ccmd.Parameters.AddWithValue("@email", c.Email ??
(object)DBNull.Value);
239             ccmd.Parameters.AddWithValue("@job", c.JobTitle ??
(object)DBNull.Value);
240             ccmd.Parameters.AddWithValue("@phone", c.PhoneNumber ??
(object)DBNull.Value);
241             ccmd.Parameters.AddWithValue("@dept", c.Department ??
(object)DBNull.Value);
242             ccmd.ExecuteNonQuery();
243
244             using var link = _conn.CreateCommand();
245             link.CommandText = "INSERT IGNORE INTO internship_contacts
(internship_id, contact_id) VALUES (@iid, @cid)";
246             link.Parameters.AddWithValue("@iid", internship.Id);
247             link.Parameters.AddWithValue("@cid", c.Id);
248             link.ExecuteNonQuery();
249         }
250     }
```

```

251
252     public IEnumerable<Internship> GetInternshipsOverview(Period period,
253     InternshipCategory? category = null)
254     {
255         var list = new List<Internship>();
256         using var cmd = _conn!.CreateCommand();
257         cmd.CommandText = @"SELECT i.id, i.organization_id, o.name as orname,
258 i.title, i.short_description, i.long_description, i.year, i.semester, i.category,
259 i.is_completed, i.final_grade
260 FROM internships i
261 LEFT JOIN organizations o ON o.organization_id = i.organization_id
262 WHERE i.year=@year AND i.semester=@sem";
263         cmd.Parameters.AddWithValue("@year", period.Year);
264         cmd.Parameters.AddWithValue("@sem", period.Semester);
265         using var rdr = cmd.ExecuteReader();
266         while (rdr.Read())
267         {
268             var cat = (InternshipCategory)Convert.ToInt32(rdr["category"]);
269             if (category != null && cat != category) continue;
270
271             var intern = new Internship(
272                 rdr.GetString("organization_id"),
273                 rdr["orgname"]?.ToString() ?? "",
274                 rdr["title"]?.ToString() ?? "",
275                 rdr["short_description"] == DBNull.Value ? "" :
276                 rdr.GetString("short_description"),
277                 rdr["long_description"] == DBNull.Value ? "" :
278                 rdr.GetString("long_description"),
279                 new Period(Convert.ToInt32(rdr["year"]),
280                 Convert.ToInt32(rdr["semester"])),
281                 cat
282             );
283             intern.Id = rdr.GetString("id");
284             if (Convert.ToInt32(rdr["is_completed"]) == 1)
285             {
286                 var grade = rdr["final_grade"] == DBNull.Value ? (double?)null
287                 : Convert.ToDouble(rdr["final_grade"]);
288                 if (grade != null) intern.MarkCompleted(grade.Value);
289             }
290             list.Add(intern);
291         }
292         rdr.Close();
293
294         foreach (var intern in list)
295         {
296             intern.Contacts.AddRange(GetContactPersonsFor-
297             Internship(intern.Id));
298             intern.AssignedStudents.AddRange(GetAssignedStudentsF-
299             orInternship(intern.Id));
300         }

```

```
292
293         return list.OrderBy(i => i.OrganizationName).ToList();
294     }
295
296     public IEnumerable<ContactPerson> GetContactPersonsForInternship(string
internshipId)
297     {
298         var list = new List<ContactPerson>();
299         using var cmd = _conn!.CreateCommand();
300         cmd.CommandText = @"SELECT cp.id, cp.first_name, cp.last_name,
cp.email, cp.job_title, cp.phone, cp.department
301 FROM contact_persons cp
302 JOIN internship_contacts ic ON ic.contact_id = cp.id
303 WHERE ic.internship_id = @iid
304 ORDER BY cp.last_name, cp.first_name";
305         cmd.Parameters.AddWithValue("@iid", internshipId);
306         using var rdr = cmd.ExecuteReader();
307         while (rdr.Read())
308         {
309             var cp = new ContactPerson
310             {
311                 Id = rdr.GetString("id"),
312                 FirstName = rdr.IsDBNull("first_name") ? null :
rdr.GetString("first_name"),
313                 LastName = rdr.IsDBNull("last_name") ? null :
rdr.GetString("last_name"),
314                 Email = rdr.IsDBNull("email") ? null : rdr.GetString("email"),
315                 JobTitle = rdr.IsDBNull("job_title") ? null :
rdr.GetString("job_title"),
316                 PhoneNumber = rdr.IsDBNull("phone") ? null :
rdr.GetString("phone"),
317                 Department = rdr.IsDBNull("department") ? null :
rdr.GetString("department")
318             };
319             list.Add(cp);
320         }
321         rdr.Close();
322         return list;
323     }
324
325     private IEnumerable<Internship> GetAllInternships()
326     {
327         using var cmd = _conn!.CreateCommand();
328         cmd.CommandText = "SELECT id, organization_id, title,
short_description, long_description, year, semester, category, is_completed,
final_grade FROM internships";
329         using var rdr = cmd.ExecuteReader();
330         var list = new List<Internship>();
331         while (rdr.Read())
332         {
```



```

333         var intern = new Internship(
334             rdr.GetString("organization_id"),
335             "",
336             rdr["title"]?.ToString() ?? "",
337             rdr["short_description"] == DBNull.Value ? "" :
rdr.GetString("short_description"),
338             rdr["long_description"] == DBNull.Value ? "" :
rdr.GetString("long_description"),
339             new Period(Convert.ToInt32(rdr["year"]),
Convert.ToInt32(rdr["semester"])),
340             (InternshipCategory)Convert.ToInt32(rdr["category"]))
341         );
342         intern.Id = rdr.GetString("id");
343         list.Add(intern);
344     }
345     rdr.Close();
346     return list;
347 }
348
349 private IEnumerable<Student> GetAllStudents()
350 {
351     var list = new List<Student>();
352     using var cmd = _conn!.CreateCommand();
353     cmd.CommandText = "SELECT student_id, first_name, last_name FROM
students";
354     using var rdr = cmd.ExecuteReader();
355     while (rdr.Read())
356     {
357         var s = new Student(rdr.GetString("student_id"),
rdr.IsDBNull("first_name") ? "" : rdr.GetString("first_name"),
rdr.IsDBNull("last_name") ? "" : rdr.GetString("last_name"));
358         list.Add(s);
359     }
360     rdr.Close();
361     return list;
362 }
363
364 public bool AddStudent(Student student)
365 {
366     if (student == null || string.IsNullOrWhiteSpace(student.StudentId))
return false;
367     using var cmd = _conn!.CreateCommand();
368     cmd.CommandText = @"
369 INSERT INTO students (student_id, first_name, last_name)
370 VALUES (@sid, @first, @last)
371 ON DUPLICATE KEY UPDATE first_name=VALUES(first_name),
last_name=VALUES(last_name);";
372     cmd.Parameters.AddWithValue("@sid", student.StudentId);
373     cmd.Parameters.AddWithValue("@first", student.FirstName ??
(object)DBNull.Value);

```

```
374         cmd.Parameters.AddWithValue("@last", student.LastName ??  
(object)DBNull.Value);  
375         cmd.ExecuteNonQuery();  
376         return true;  
377     }  
378  
379     public bool AssignStudentToInternship(Student student, string internshipId)  
380     {  
381         using var cmd = _conn!.CreateCommand();  
382         cmd.CommandText = "SELECT year, semester FROM internships WHERE  
id=@iid";  
383         cmd.Parameters.AddWithValue("@iid", internshipId);  
384         using var rdr = cmd.ExecuteReader();  
385         if (!rdr.Read()) { rdr.Close(); return false; }  
386         var year = Convert.ToInt32(rdr["year"]); var sem =  
Convert.ToInt32(rdr["semester"]);  
387         rdr.Close();  
388  
389         using var chk = _conn.CreateCommand();  
390         chk.CommandText = @"SELECT COUNT(*) FROM internship_assignments ia  
391 JOIN internships i ON ia.internship_id = i.id  
392 WHERE ia.student_id = @sid AND i.year = @year AND i.semester = @sem";  
393         chk.Parameters.AddWithValue("@sid", student.StudentId);  
394         chk.Parameters.AddWithValue("@year", year);  
395         chk.Parameters.AddWithValue("@sem", sem);  
396         var conflict = Convert.ToInt32(chk.ExecuteScalar() ?? 0) > 0;  
397         if (conflict) return false;  
398  
399         using var upsert = _conn.CreateCommand();  
400         upsert.CommandText = @"  
401 INSERT INTO students (student_id, first_name, last_name) VALUES (@sid,@first,@last)  
402 ON DUPLICATE KEY UPDATE first_name=VALUES(first_name),  
last_name=VALUES(last_name);";  
403         upsert.Parameters.AddWithValue("@sid", student.StudentId);  
404         upsert.Parameters.AddWithValue("@first", student.FirstName ??  
(object)DBNull.Value);  
405         upsert.Parameters.AddWithValue("@last", student.LastName ??  
(object)DBNull.Value);  
406         upsert.ExecuteNonQuery();  
407  
408         using var assign = _conn.CreateCommand();  
409         assign.CommandText = "INSERT IGNORE INTO internship_assignments  
(internship_id, student_id) VALUES (@iid, @sid);";  
410         assign.Parameters.AddWithValue("@iid", internshipId);  
411         assign.Parameters.AddWithValue("@sid", student.StudentId);  
412         assign.ExecuteNonQuery();  
413  
414         return true;  
415     }  
416
```

```
417     public bool RemoveStudentFromInternship(string studentId, string
internshipId)
418     {
419         using var cmd = _conn!.CreateCommand();
420         cmd.CommandText = "DELETE FROM internship_assignments WHERE
internship_id=@iid AND student_id=@sid";
421         cmd.Parameters.AddWithValue("@iid", internshipId);
422         cmd.Parameters.AddWithValue("@sid", studentId);
423         var affected = cmd.ExecuteNonQuery();
424         return affected > 0;
425     }
426
427     public bool WithdrawInternship(string internshipId)
428     {
429         if (string.IsNullOrEmpty(internshipId)) return false;
430
431         using var tran = _conn!.BeginTransaction();
432         try
433         {
434             string? orgId = null;
435             using (var getOrg = _conn.CreateCommand())
436             {
437                 getOrg.Transaction = tran;
438                 getOrg.CommandText = "SELECT organization_id FROM internships
WHERE id=@iid";
439                 getOrg.Parameters.AddWithValue("@iid", internshipId);
440                 var o = getOrg.ExecuteScalar();
441                 orgId = o == null || o == DBNull.Value ? null : o.ToString();
442             }
443
444             using (var delAssign = _conn.CreateCommand())
445             {
446                 delAssign.Transaction = tran;
447                 delAssign.CommandText = "DELETE FROM internship_assignments
WHERE internship_id=@iid";
448                 delAssign.Parameters.AddWithValue("@iid", internshipId);
449                 delAssign.ExecuteNonQuery();
450             }
451
452             using (var delIC = _conn.CreateCommand())
453             {
454                 delIC.Transaction = tran;
455                 delIC.CommandText = "DELETE FROM internship_contacts WHERE
internship_id=@iid";
456                 delIC.Parameters.AddWithValue("@iid", internshipId);
457                 delIC.ExecuteNonQuery();
458             }
459
460             int affected;
461             using (var delIntern = _conn.CreateCommand())
```

```
462         {
463             delIntern.Transaction = tran;
464             delIntern.CommandText = "DELETE FROM internships WHERE
id=@iid";
465             delIntern.Parameters.AddWithValue("@iid", internshipId);
466             affected = delIntern.ExecuteNonQuery();
467         }
468
469         if (!string.IsNullOrEmpty(orgId))
470         {
471             using var delOrphanContacts = _conn.CreateCommand();
472             delOrphanContacts.Transaction = tran;
473             delOrphanContacts.CommandText =
474                 @"DELETE FROM contact_persons
475                 WHERE organization_id=@org
476                 AND id NOT IN (SELECT contact_id FROM
internship_contacts)";
477             delOrphanContacts.Parameters.AddWithValue("@org", orgId);
478             delOrphanContacts.ExecuteNonQuery();
479         }
480
481         tran.Commit();
482         return affected > 0;
483     }
484     catch
485     {
486         try { tran.Rollback(); } catch { }
487         return false;
488     }
489 }
490
491 public bool MarkInternshipCompleted(string internshipId, double finalGrade)
492 {
493     using var cmd = _conn!.CreateCommand();
494     cmd.CommandText = "UPDATE internships SET is_completed=1,
final_grade=@grade WHERE id=@iid";
495     cmd.Parameters.AddWithValue("@grade", finalGrade);
496     cmd.Parameters.AddWithValue("@iid", internshipId);
497     var affected = cmd.ExecuteNonQuery();
498     return affected > 0;
499 }
500
501 public Organization? GetOrganizationById(string organizationId)
502 {
503     using var cmd = _conn!.CreateCommand();
504     cmd.CommandText = "SELECT organization_id, name, address, city, url,
phone, email FROM organizations WHERE organization_id=@id";
505     cmd.Parameters.AddWithValue("@id", organizationId);
506     using var rdr = cmd.ExecuteReader();
507     if (!rdr.Read()) { rdr.Close(); return null; }
```

```

508         var org = new Organization(
509             rdr.GetString("organization_id"),
510             rdr.GetString("name"),
511             rdr.IsDBNull("address") ? "" : rdr.GetString("address"),
512             rdr.IsDBNull("city") ? "" : rdr.GetString("city")
513         )
514         {
515             Url = rdr.IsDBNull("url") ? null : rdr.GetString("url"),
516             PhoneNumber = rdr.IsDBNull("phone") ? null :
rdr.GetString("phone"),
517             Email = rdr.IsDBNull("email") ? null : rdr.GetString("email")
518         };
519         rdr.Close();
520
521         using var ccmd = _conn.CreateCommand();
522         ccmd.CommandText = "SELECT id, first_name, last_name, email, job_title,
phone, department FROM contact_persons WHERE organization_id=@org ORDER BY
last_name, first_name";
523         ccmd.Parameters.AddWithValue("@org", organizationId);
524         using var cr = ccmd.ExecuteReader();
525         while (cr.Read())
526         {
527             var cp = new ContactPerson
528             {
529                 Id = cr.GetString("id"),
530                 FirstName = cr.IsDBNull("first_name") ? null :
cr.GetString("first_name"),
531                 LastName = cr.IsDBNull("last_name") ? null :
cr.GetString("last_name"),
532                 Email = cr.IsDBNull("email") ? null : cr.GetString("email"),
533                 JobTitle = cr.IsDBNull("job_title") ? null :
cr.GetString("job_title"),
534                 PhoneNumber = cr.IsDBNull("phone") ? null :
cr.GetString("phone"),
535                 Department = cr.IsDBNull("department") ? null :
cr.GetString("department")
536             };
537             org.Contacts.Add(cp);
538         }
539         cr.Close();
540
541         return org;
542     }
543
544     public IEnumerable<Internship> GetInternshipsByOrganizationId(string
organizationId)
545     {
546         using var cmd = _conn!.CreateCommand();
547         cmd.CommandText = @"SELECT id, title, short_description,
long_description, year, semester, category, is_completed, final_grade

```

```
548 FROM internships WHERE organization_id=@org ORDER BY year, semester";
549 cmd.Parameters.AddWithValue("@org", organizationId);
550 using var rdr = cmd.ExecuteReader();
551 var list = new List<Internship>();
552 while (rdr.Read())
553 {
554     var intern = new Internship(
555         organizationId,
556         "",
557         rdr["title"]?.ToString() ?? "",
558         rdr["short_description"] == DBNull.Value ? "" :
rdr.GetString("short_description"),
559         rdr["long_description"] == DBNull.Value ? "" :
rdr.GetString("long_description"),
560         new Period(Convert.ToInt32(rdr["year"]),
Convert.ToInt32(rdr["semester"])),
561         (InternshipCategory)Convert.ToInt32(rdr["category"]))
562     );
563     intern.Id = rdr.GetString("id");
564     list.Add(intern);
565 }
566 rdr.Close();
567 return list;
568 }
569
570 private IEnumerable<Student> GetAssignedStudentsForInternship(string
internshipId)
571 {
572     var list = new List<Student>();
573     using var cmd = _conn!.CreateCommand();
574     cmd.CommandText = @"SELECT s.student_id, s.first_name, s.last_name
575 FROM students s
576 JOIN internship_assignments ia ON ia.student_id = s.student_id
577 WHERE ia.internship_id = @iid";
578     cmd.Parameters.AddWithValue("@iid", internshipId);
579     using var rdr = cmd.ExecuteReader();
580     while (rdr.Read())
581     {
582         list.Add(new Student(rdr.GetString("student_id"),
rdr.IsDBNull("first_name") ? "" : rdr.GetString("first_name"),
rdr.IsDBNull("last_name") ? "" : rdr.GetString("last_name")));
583     }
584     rdr.Close();
585     return list;
586 }
587
588 public bool AddContactPerson(string organizationId, ContactPerson contact)
589 {
590     if (contact == null || string.IsNullOrWhiteSpace(organizationId))
return false;
```

```

591         using var cmd = _conn!.CreateCommand();
592         cmd.CommandText = @"
593 INSERT INTO contact_persons (id, organization_id, first_name, last_name, email,
594 job_title, phone, department)
595 VALUES (@id,@org,@first,@last,@email,@job,@phone,@dept)
596 ON DUPLICATE KEY UPDATE first_name=VALUES(first_name), last_name=VALUES(last_name),
597 email=VALUES(email), job_title=VALUES(job_title), phone=VALUES(phone),
598 department=VALUES(department), organization_id=VALUES(organization_id);";
599         cmd.Parameters.AddWithValue("@id",
600 string.IsNullOrEmpty(contact.Id) ? Guid.NewGuid().ToString() : contact.Id);
601         cmd.Parameters.AddWithValue("@org", organizationId);
602         cmd.Parameters.AddWithValue("@first", contact.FirstName ??
603 (object)DBNull.Value);
604         cmd.Parameters.AddWithValue("@last", contact.LastName ??
605 (object)DBNull.Value);
606         cmd.Parameters.AddWithValue("@email", contact.Email ??
607 (object)DBNull.Value);
608         cmd.Parameters.AddWithValue("@job", contact.JobTitle ??
609 (object)DBNull.Value);
610         cmd.Parameters.AddWithValue("@phone", contact.PhoneNumber ??
611 (object)DBNull.Value);
612         cmd.Parameters.AddWithValue("@dept", contact.Department ??
613 (object)DBNull.Value);
614         cmd.ExecuteNonQuery();
615         return true;
616     }
617
618     public IEnumerable<ContactPerson> GetContactPersonsByOrganization(string
619 organizationId)
620     {
621         var list = new List<ContactPerson>();
622         using var cmd = _conn!.CreateCommand();
623         cmd.CommandText = "SELECT id, first_name, last_name, email, job_title,
624 phone, department FROM contact_persons WHERE organization_id=@org ORDER BY
625 last_name, first_name";
626         cmd.Parameters.AddWithValue("@org", organizationId);
627         using var rdr = cmd.ExecuteReader();
628         while (rdr.Read())
629         {
630             var cp = new ContactPerson
631             {
632                 Id = rdr.IsDBNull("id") ? Guid.NewGuid().ToString() :
633 rdr.GetString("id"),
634                 FirstName = rdr.IsDBNull("first_name") ? null :
635 rdr.GetString("first_name"),
636                 LastName = rdr.IsDBNull("last_name") ? null :
637 rdr.GetString("last_name"),
638                 Email = rdr.IsDBNull("email") ? null : rdr.GetString("email"),
639                 JobTitle = rdr.IsDBNull("job_title") ? null :
640 rdr.GetString("job_title"),

```

```
624         PhoneNumber = rdr.IsDBNull("phone") ? null :  
rdr.GetString("phone"),  
625         Department = rdr.IsDBNull("department") ? null :  
rdr.GetString("department")  
626     };  
627     list.Add(cp);  
628 }  
629 rdr.Close();  
630 return list;  
631 }  
632  
633 public void Dispose()  
634 {  
635     if (_conn != null) { try { _conn.Close(); } catch { } _conn = null; }  
636 }  
637 }  
638 }
```